

0/1 Knapsack Problem

- Knapsack - bag
- 0/1 Knapsack means that items are either completely or no items are filled in a knapsack.
- * Objective - to place the object into knapsack so that the profit is maximum.
 - * We have to select the items in such a way that the sum of weight of items should be either smaller than or equal to the weight of the container and the profit should be maximum.
- n objects, each object - Profit P_i
Weight w_i
Knapsack capacity = m

Example : $n=4$ and $m=15$

$$P_i = (10, 10, 12, 18) \text{ and } w_i = (2, 4, 6, 9)$$

first, convert all profits into negatives

$$P_i = (-10, -10, -12, 18)$$

for Node ①, calculate upper bound and lower bound

- * upper bound fraction values are not allowed
- * In lower bound fraction values are allowed.

Node 1

Calculate lower bound

-18	3/9	(3)
-12	6	(9)
-10	4	(13)
-10	2	

$$= -10 - 10 - 12 - 18 \times \frac{3}{9}$$

$$= -32 - 6 = -38$$

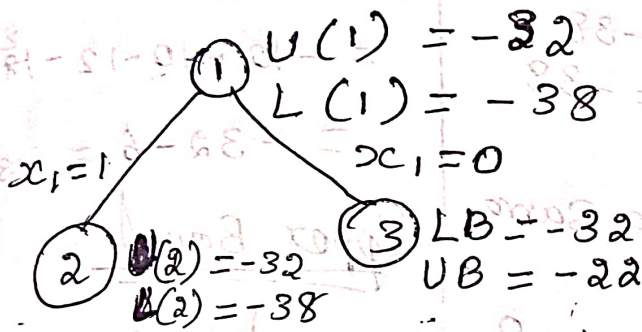
$$LB = -38$$

Upper Bound

-12	6	(9)
-10	4	(13)
-10	2	

$$-10 - 10 - 12 = -32$$

$$UB = -32$$



Node 2

Lower

-18	3/4	(3)
-12	6	(9)
-10	4	(13)

$$\Rightarrow -10 - 10 - 12 - 18 \times \frac{3}{4}$$

$$= -32 - 6 = -38$$

Upper

		(3)
6		(9)
4		(13)
2		

$$= -10 - 10 - 12 = -32$$

Node 3

Lower $x_1 = 0$

-18	5/4	(3)
-12	6	(9)
-10	4	(13)

$$= -10 - 12 - 18 \times \frac{5}{4}$$

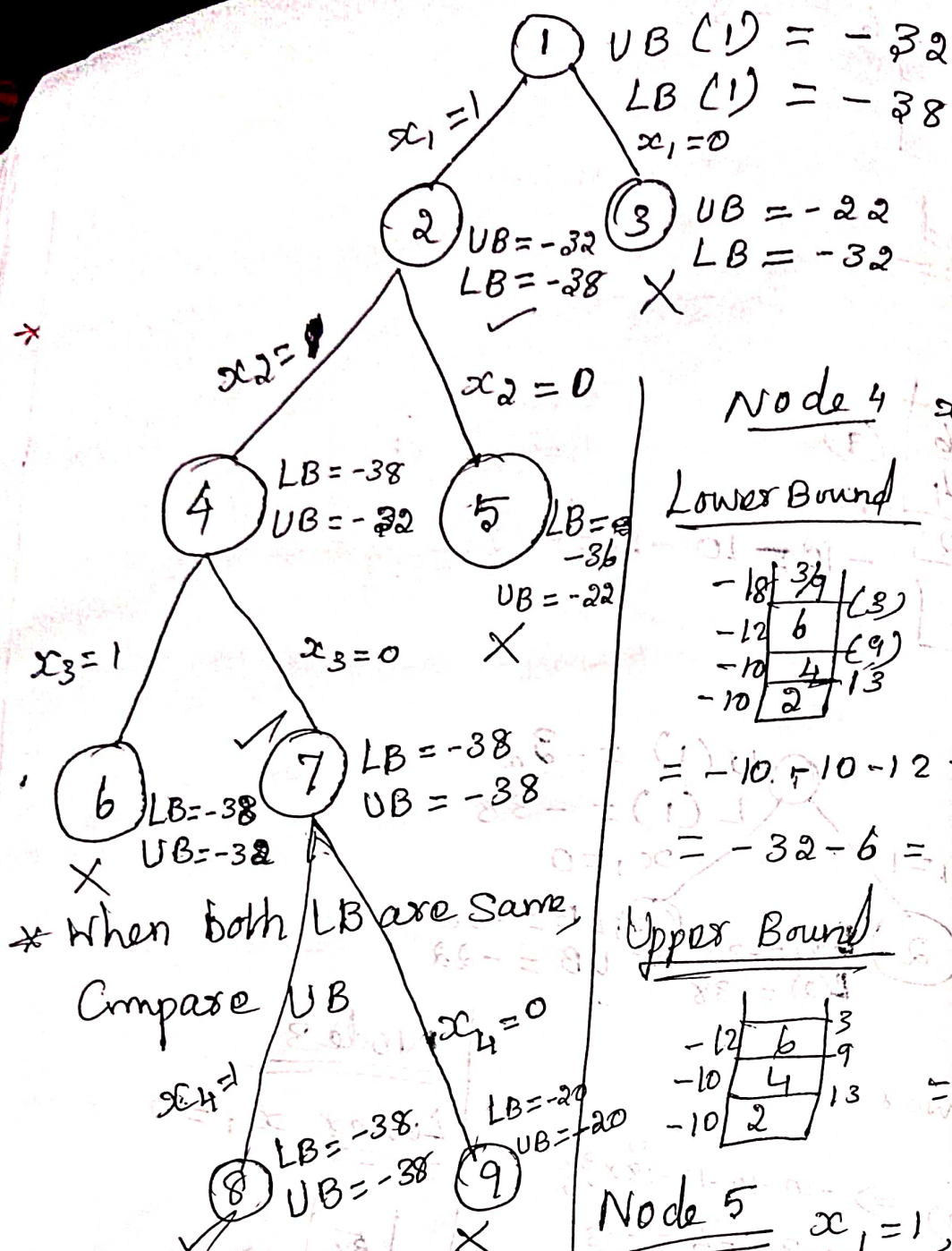
$$= -10 - 12 - 10 = -32$$

Upper

$x_1 = 0$

-12	6	
-10	4	

$$= -22$$



* When both LB are same, Compare UB

$x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 1$

Total profit = $-10 - 10 - 18$
 $= -38$
 Nothing Better
 $+38$

Node 4 $x_1 = 1, x_2 = 0$
Lower Bound

-18	3/9	(3)
-12	6	(9)
-10	4	13
-10	2	

$= -10 - 10 - 12 - 18 \times \frac{3}{9}$
 $= -32 - 6 = -38$

Upper Bound

-12	6	9
-10	4	13
-10	2	

$= -32$

Node 5 $x_1 = 1, x_2 = 0$
Lower Bound

-18	7/9	7
-12	6	13
-10	2	

$= -10 - 12 - 18 \times \frac{7}{9}$
 $= -36$

Upper Bound

-12	6	
-10	2	

$= -22$

Node 6 $x_1=1, x_2=1, x_3=1$

Upper Bound

$$UB = -32$$

Lower Bound

$$LB = -38$$

~~Lower Bound~~

Node 7 $x_1=1, x_2=1, x_3=0$

Lower Bound

-18	9	(0)
-10	4	(9)
-10	2	(13)

$$= -38$$

Upper Bound

$$UB = -38$$

Node 8 $x_1=1, x_2=1, x_3=0$

LB

$$x_4=1$$

-18	9	0
-10	4	9
-10	2	13

$$LB = -38$$

UB

$$UB = -38$$

Node 9

$x_1=1, x_2=1, x_3=0, x_4=0$

UB

$$UB = -20$$

LB

-18	9	0
-10	4	9
-10	2	13

$$= -20$$